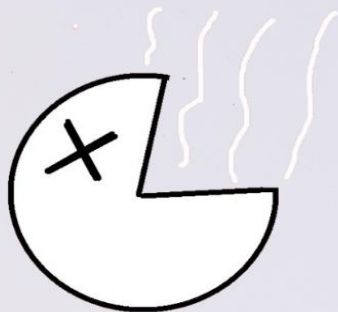




TOTOZ COLLECTION

SKIN PATHOLOGY
[PETRENKOZ EDITION]



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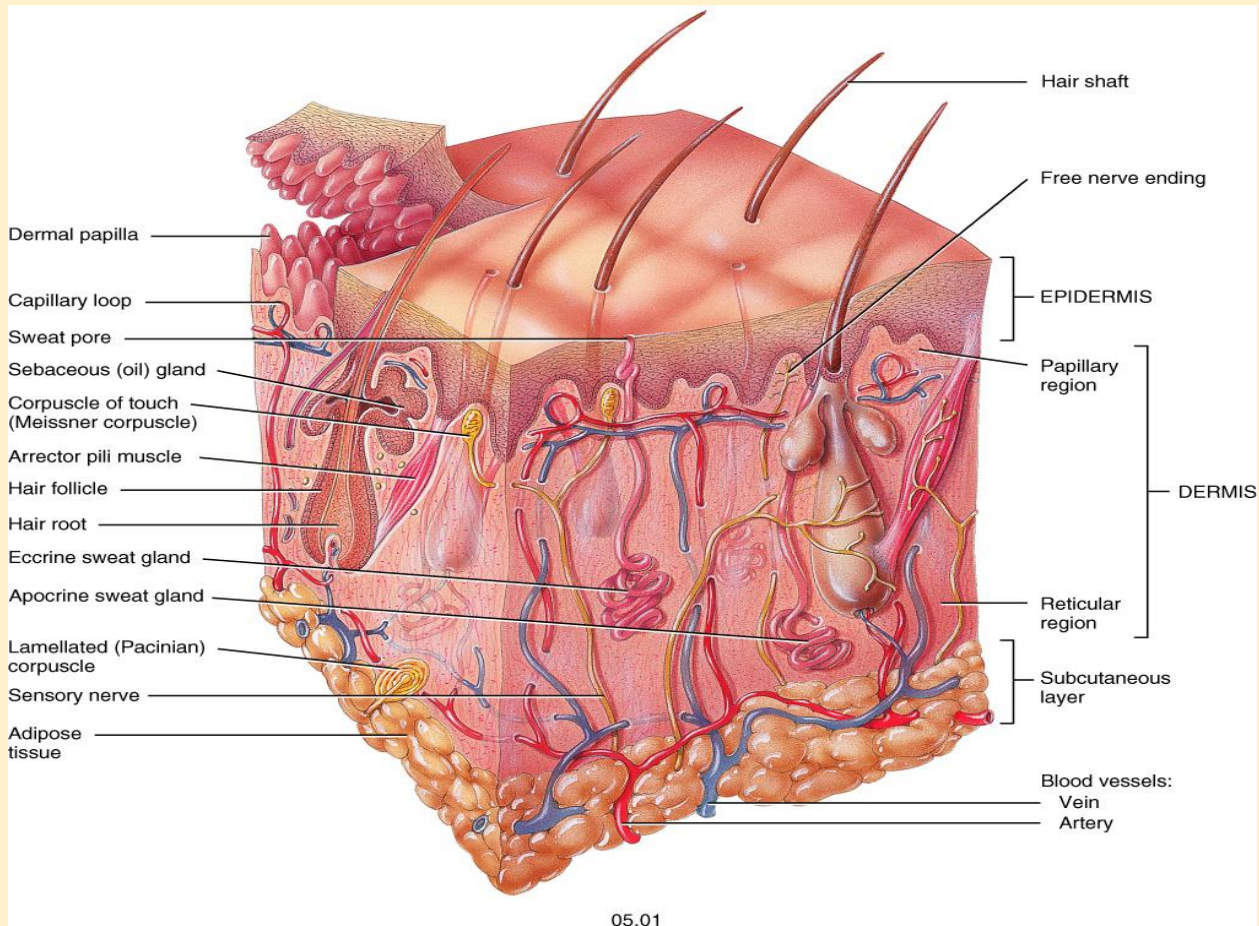
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SKIN



1. INTRODUCTION

- The two major layers of the skin-the superficial **epidermis** and deeper **dermis** -are derived from different embryonic components and retain a radically different morphology.
- The **dermis** is relatively acellular and is recognizably divided into two zones:
 - ✓ **papillary dermis** ; 20% dermis dermal papilla. Vascular Layer
 - ✓ **reticular dermis**. 80% Collagen and Elastin fibre
- Both regions of the dermis contain blood and lymph vessels as well as nerves.
- The **epidermis** is a multilayered sheet of keratin-producing cells.

- A progressive change in morphology occurs from the replicating columnar cells of the basal layer (**stratum basalis**) through the spinous layer (**stratum spinosum**) and the granular layer (**stratum granulosum**) to the nonviable flattened cells of the cornified layer (**stratum corneum**)
- Accessory Cells of Epidermis
 - ✓ **Langerhan Cells** – Defence cells macrophages
 - ✓ **Melanocytes**- Give colour to skin- neural crest cells (Albinos)
 - ✓ **Merkels Cells**- Nerves cells sensation
 - ✓ **Keratinocytes Cells** of the epidermis
- Two types of cells, melanocytes and Langerhans cells, are dendritic. The third, the Merkel cell, is associated with a terminal neuronal axon.

2. FUNCIONS OF THE SKIN

- The skin is a protective barrier; microorganisms find it almost impossible to penetrate the epidermis from the outside, and
- water loss is limited from the inside.
- The skin is vital in regulating temperature and protecting against ultraviolet light.
- A variety of sensory receptors communicate details related to the immediate environment.
- The skin plays a prominent role in immune regulation, where you have the Keratinocytes, Langerhans cells, mast cells, lymphocytes and macrophages all serve functions related to immunity.

3. FEATURES OF SKIN LESIONS

- Dermatology is a visual discipline, and it relies on accurate macroscopic and microscopic description of skin lesions. Thus it is important to pay attention to the basic characteristics of each lesion, including its:
 - ↪ Size
 - ↪ Color
 - ↪ Location and distribution
 - ↪ Consistency
 - ↪ Shape

4. SKIN LESIONS CLASSIFIED

- In clinical practice, it is customary to classify skin lesions as **primary** or **secondary**.
- Primary skin lesions include **macules**, **papules**, **patches**, **plaques**, **nodules**, **cysts**, wheals, **vesicles**, **bullae**, and **pustules**.
- Secondary skin lesions are related to the progression of the disease but may also be caused by scratching, trauma, or treatment. This group of lesions includes **crusts**, **erosions**, **ulcers**, **fissures**, **excoriations**, **scars**, and **scales**.

5. DEFINITION OF TERMS

TERM	DESCRIPTION
Acanthosis	⊗ A histopathologic term describing thickening of the epidermis; may be regular or irregular. ⊗ Thickening of the epidermis due to hyperplasia of stratum malpighii. ⊗ Epidermal hyperplasia preferentially involving the stratum spinosum.
Acantholysis	⊗ Loss of intercellular connections resulting in lack of cohesion between keratinocytes ⊗ Loss of cohesion between epidermal cells with formation of intraepidermal space containing oedema fluid and detached epithelial cells.
Dyskeratosis	⊗ Abnormal development of epidermal cells resulting in rounded cells devoid of their prickles and having pyknotic nuclei. Dyskeratosis is a feature of premalignant and malignant lesions and is rarely seen in benign conditions.
Spongiosis	⊗ A histopathologic term describing the separation of individual epidermal keratinocytes by intercellular edema. ⊗ Intercellular edema of the epidermis.
Hyperkeratosis	⊗ Hyperplasia of the stratum corneum, often associated with a qualitative abnormality of the keratin.

	⊗ Thickening of the horny layer.
Parakeratosis	⊗ A histopathologic term describing the retention of nuclei within the stratum corneum. ⊗ Abnormal keratinisation of the cells so that the horny layer contains nucleated keratinocytes rather than the normal non-nucleate keratin layer. ⊗ Mode(s) of keratinization characterized by retention of the nuclei in the stratum corneum; on mucosal membranes, parakeratosis is normal.
Papillomatosis	⊗ Hyperplasia of the papillary dermis with elongation and/or widening of the dermal papillae.
Veruccae vulgaris (vv).	⊗ Groups of large vacuolated cells lie in the upper stratum malpighii and in the granular layer. The thickened horny layer contains many parakeratotic cells.
Psoriasiform	⊗ A histopathologic term describing uniform, regular thickening of the epidermis as seen in classic psoriasis; regular acanthosis
Pigment incontinence	⊗ Loss of melanin pigment from damaged basal cell layer so that the pigment accumulates in the melanophages in the dermis.
Dermatosis	⊗ is a common term used for any skin disorder. ⊗ may be of various types such as genetic, inflammatory, infectious, granulomatous, connective tissue, bullous and scaling type.

6. HYPERSENSITIVITY REACTIONS

- Four hypersensitivity reactions are generally recognized, and all can affect the skin.
- Type I hypersensitivity reaction, or atopic reaction,
 - ⊗ is based on the reaction of antigens with the immunoglobulin (Ig) E bound to the surface of mast cells in the skin. Degranulation of mast cells leads to the formation of wheals (hives), which result from the increased permeability of blood vessels induced by histamine. Bee stings, allergic reactions to drugs, and atopic

dermatitis of childhood are other examples of type I hypersensitivity.

- **Type II hypersensitivity reaction, or cytotoxic cell reaction,**
 - ⊗ is mediated by antibodies that attach to antigens on the surface of epidermal cells (as in pemphigus vulgaris) or the epidermal-dermal basement membrane (as in bullous pemphigoid). Type II hypersensitivity reaction leads to the formation of bullae.
- **Type III hypersensitivity reaction**
 - ⊗ is based on the deposition of preformed circulating antigen-antibody complexes along the epidermal-dermal basement membrane (as in SLE) or in the vessel wall (as in hypersensitivity vasculitis).
- **Type IV hypersensitivity reaction**
 - ⊗ is based on the reaction of T lymphocytes and is typically induced by foreign antigens, as in contact dermatitis (e.g., poison ivy). It may be accompanied by the formation of granulomas, as in hypersensitivity to *Mycobacterium tuberculosis* or *Mycobacterium leprae*. Sarcoidosis is a type IV hypersensitivity reaction to unknown antigens.

7. CLASSIFICATION OF SKIN DISEASES

Inflammatory Diseases of the Superficial and Deep Vascular Bed

- *Urticaria and Angioedema*
- *Cutaneous Necrotizing Vasculitis*
- *Allergic Contact Dermatitis*
- *Granulomatous Dermatitis*
- *Sarcoidosis*
- *Granuloma Annulare*

Diseases of the Basement Membrane Zone

- *Epidermolysis Bullosa*
- *Bullous Pemphigoid*
- *Dermatitis Herpetiformis*
- *Erythema Multiforme*
- *Systemic Lupus Erythematosus*
- *Lichen Planus*

Diseases of the Epidermis

- *Ichthyoses*
- *Darier Disease*
- *Psoriasis*

- **Disorder of the Dermal Connective Tissue**
 - **Scleroderma**
- **Inflammatory Disorders of the Panniculus**
 - **Erythema Nodosum**
 - **Erythema Induratum**
- **Disorder of the Pilosebaceous Unit**
 - **Acne Vulgaris**

Infections and Infestations

- **Impetigo**
- **Superficial Fungal Infections**
- **Deep Fungal Infections**
- **Viral Infections**
- **Arthropod Infestations**

- **Primary Neoplasms of the Skin**
 - **Common Acquired Melanocytic Nevus (Mole)**
 - **Dysplastic (Atypical) Nevus**
 - **Melanocytic Dysplasia**
 - **Malignant Melanoma**
 - **Benign Tumors of Melanocytes**
 - **Verrucae**
 - **Keratosis**
 - **Basal Cell Carcinoma**
 - **Squamous Cell Carcinoma**
 - **Merkel Cell Carcinoma**
 - **Adnexal Tumors**
 - **Fibrohistiocytic Tumors**
 - **Mycosis Fungoides**

8. SCALING DERMATOSES

8.1. ICHTHYOSIFORM DERMATOSES

8.1.1. INTRODUCTIN

- Ichthyosiform dermatoses, many of which are heritable, are a heterogeneous group of diseases characterized by striking thickening of the stratum corneum.

- The term ichthyosis reflects the similarity of the diseased skin to coarse, fish-like scales. Several rare ichthyoses are associated with other abnormalities such as abnormal lipid metabolism, neurologic disorders, bone diseases and cancer.

8.1.2. PATHOGENESIS

- Three general defects are involved in the excessive epidermal cornification of the ichthyoses:
 - a) **Increased cohesiveness of the cells of the stratum corneum**, possibly related to altered lipid metabolism
 - b) **Abnormal keratinization**, manifested as impaired tonofilament formation and keratohyaline synthesis and as excessive cornification
 - c) **Increased basal cell proliferation**, associated with a decrease in transit time of keratinocytes across the epidermis.

Ichthyosis vulgaris. Noninflammatory fish-like scales are evident on the thigh of a patient with a strong family history of ichthyosis vulgaris.



8.2. Psoriasis

8.2.1. Pathogenesis

- ☒ The Pathogenesis of psoriasis is poorly understood and is likely multifactorial.

8.2.1.1. Genetic Factors

- ✓ Psoriasis unquestionably has a genetic component, although only one third of patients with psoriasis have a family history of the disease. The more severe the illness, the greater the likelihood is of a familial background.

8.2.1.2. Immunologic Factors

- ✓ T lymphocytes are now considered key to the pathogenesis of psoriatic lesions.
- ✓ Eruption of psoriatic lesions coincides with T-cell infiltration into the epidermis. By contrast, resolution of psoriatic plaques, whether spontaneous or induced by treatment, follows disappearance, or reduction in, epidermal T cells.

8.2.1.3. Environmental Factors

- ✓ Clinical lesions may occur anywhere on the skin. In this context, a variety of stimuli, such as physical injury infection, certain drugs and photosensitivity, may produce psoriatic lesions in apparently normal skin.
- ✓ The pathogenesis of psoriatic plaques may be appreciated by contrasting the effect of chronic cutaneous trauma in persons with and without psoriasis.
- ✓ Chronic irritation of a normal person's skin, as in repeated rubbing, produces a tough, scaly, cutaneous plaque that is both clinically and histologically psoriasiform.

8.2.2. Microcirculatory Changes.

- ✓ In psoriatic skin, the capillary loops of the dermal papillae become venular, showing multiple layers of basal lamina material, wide lumina and "bridged" fenestrations between endothelial cells.

- ✓ The vascular change, which occurs in concert with a striking increase in neutrophilic chemotactic factors, leads to diapedesis of many neutrophils at the tips of dermal papillae and subsequent migration into the epidermis (the “squirting papillae”).
- ✓ This unusual pattern of neutrophilic inflammation is responsible for the dense collections of neutrophils in the stratum corneum (Munro microabscesses) as well as for the scattering of neutrophils throughout the epidermis (spongiform pustules of Kogoj).

8.2.2. Histological characteristics

- ☒ Focal Parakeratosis of the epidermis
- ☒ Acanthosis, with regular downgrowth of rete ridges to almost the same dermal level with thickening of their lower portion.
- ☒ Prolongation of the Rete ridges/pegs
- ☒ Exudation of neutrophils (i.e. inflammatory cells) into the dermal PAPILLAE + Fluid from the dilated thickened capillaries on the tips of the papillae.
- ☒ This leads to formation of microabscesses = in the epidermis called as MUNRO’s MICROABSCESSES
- ☒ Genetically determined disease
- ☒ Silvery-grey scales of parakeratosis
- ☒ Polymorphs enter epidermis but abscesses are sterile
- ☒ Elongation and oedema of the dermal papillae with broadening of their tips.
- ☒ Suprapapillary thinning of stratum malpighii.
- ☒ Absence of granular cell layer.
- ☒ Prominent parakeratosis.
- ☒ Presence of Munro microabscesses in the parakeratotic horny layer is diagnostic of psoriasis.

Lichen planus

Also called: lichen ruber planus

An inflammatory condition of the skin and mucous membranes.

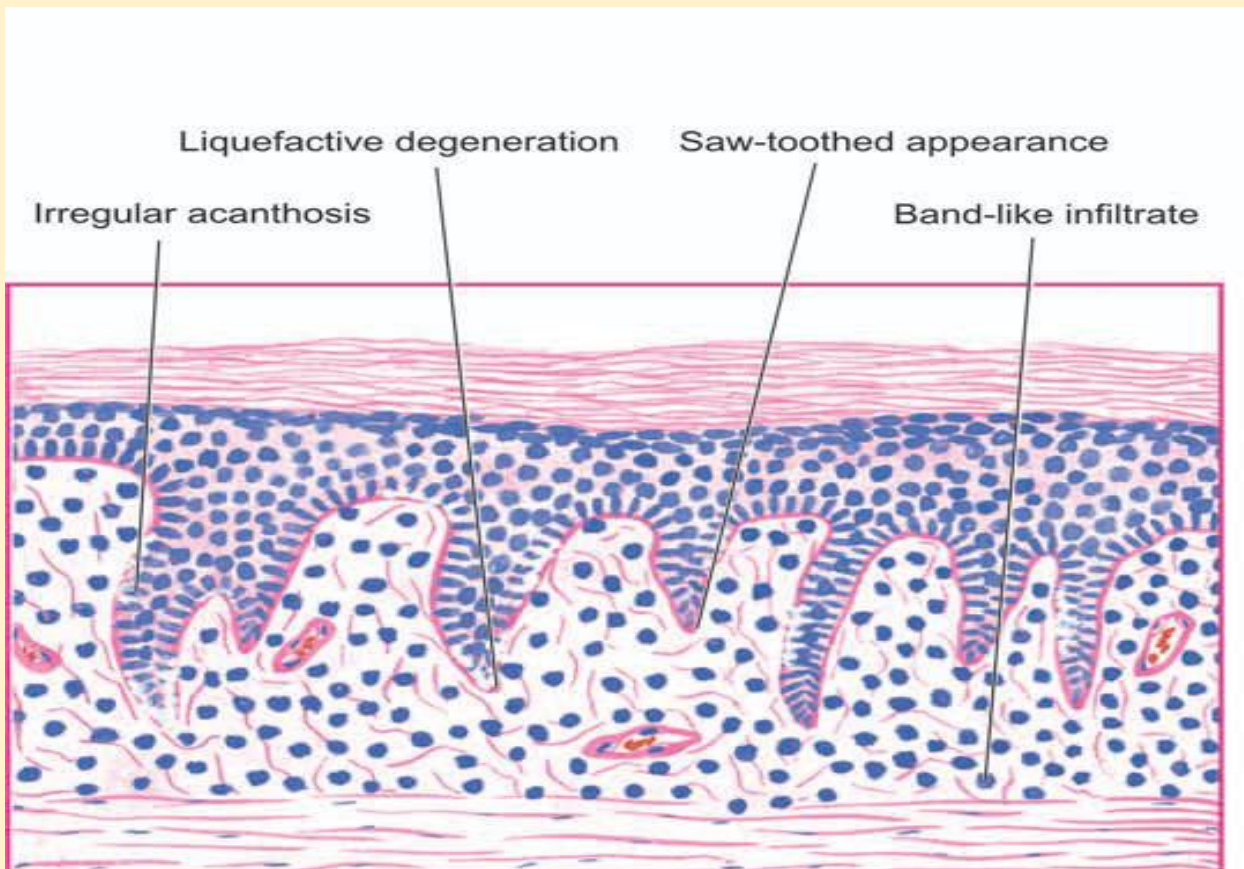
Lichen planus occurs when the immune system mistakenly attacks cells of the skin or mucous membranes.

8.3.1. Introduction

- ✚ Lichen Planus Is a Cell-Mediated Immunologic Reaction at the Dermal–Epidermal Junction
- ✚ Lichen Planus is a Hypersensitivity Reaction with Lymphocytic Infiltrates at the Dermal-Epidermal Junction
- ✚ “Lichenoid” tissue reactions are so named because the clinical lesions resemble certain lichens that form a scaly growth on rocks or tree trunks.
- ✚ Histologically, lichenoid infiltrates are characterized by a band-like congregation of lymphocytes that obscures the dermal–epidermal junction. Epidermal turnover is decreased, leading to hyperkeratosis without parakeratosis.
- ✚ The distinctive pathologic changes of LP are at the dermal–epidermal interface. The basal row of cuboidal cells is replaced by flattened or polygonal keratinocytes. The undulating interface between the dermal papillae and the rounded profiles of the rete ridges is obscured by a dense infiltrate of

lymphocytes and macrophages, many of the latter containing melanin pigment (melanophages).

- ↳ Sharply pointed (“saw-toothed”) rete ridges of keratinocytes project into the inflammatory infiltrate. Commonly admixed with the infiltrate (in the epidermis or dermis) are globular, fibrillary, eosinophilic bodies, 15 to 20 m in diameter, which represent apoptotic keratinocytes. These structures are variably termed apoptotic, colloid, Civatte or fibrillary bodies. The fibrils within the apoptotic bodies are keratin filaments. Epidermal Langerhans cells are increased early in LP.



8.3.2. Etiologic Factors

- ↳ Pathogenetic mechanisms are outlined. The disease is apparently initiated by epidermal injury. This injury causes some epidermal cells to be treated as “foreign.” The antigens of such cells are processed by Langerhans cells. The processed antigen induces lymphocytic proliferation and macrophage activation. Macrophages, along with T lymphocytes, kill the epidermal basal cells, resulting in a reactive epidermal proliferation and the formation of fibrillary bodies.

8.3.3. Clinical Presentation

- ↳ Lichen planus is a chronic dermatosis characterised clinically by irregular, violaceous, shining, flattopped, pruritic papules.
- ↳ The lesions are distributed symmetrically with sites of predilection being flexor surfaces of the wrists, forearms, legs and external genitalia. Buccal mucosa is also involved in many cases of lichen planus.
- ↳ Chronic dermatosis with skin, hair follicle, and mucous membrane involvement; 10% of cases show involvement of the nail unit, often with pterygium formation
- ↳ Presents clinically as pruritic, violaceous, flat-topped, polygonal papules, and plaques
- ↳ Sites of predilection include inner aspects of wrists, flexor forearms, anterior thighs and the shins; lichen planopilaris (follicular lichen planus) primarily affects the scalp B There is a reported association with hepatitis C seropositivity

8.3.4. Histologic Characteristics

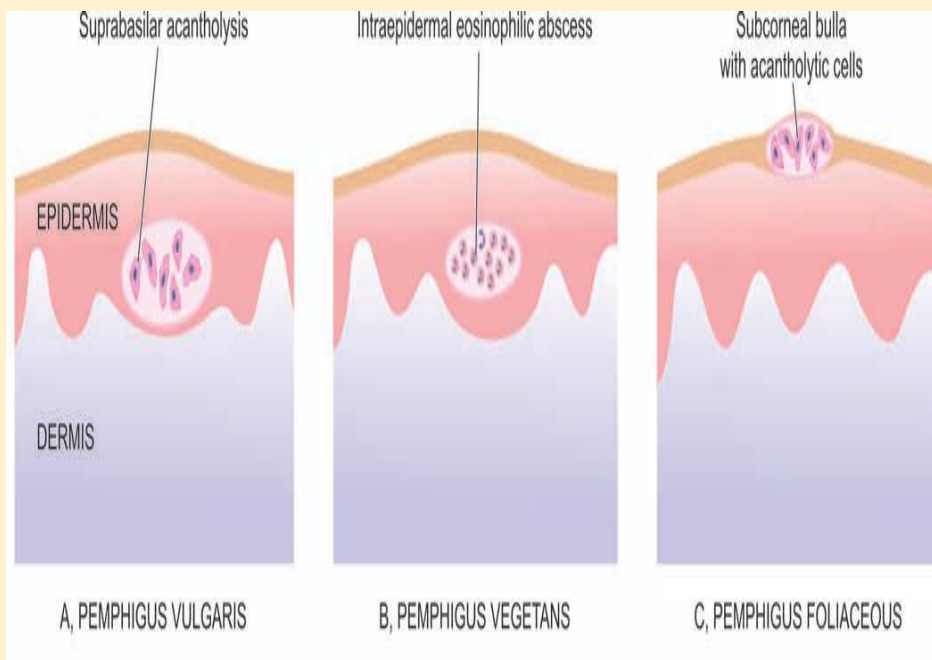
- ↳ Histologically, the characteristic features are:
 - i. Marked hyperkeratosis.
 - ii. Focal hypergranulosis.
 - iii. Irregular acanthosis with elongated saw-toothed rete ridges.
 - iv. Liquefactive degeneration of the basal layer.
 - v. A band-like dermal infiltrate of mononuclear cells, sharply demarcated at its lower border and closely hugging the basal layer.

9. Pemphigus / Bullous Skin Disorders

9.1. Introduction

- ☒ rare autoimmune blistering disorder resulting from loss of integrity of normal intercellular attachments within the epidermis and mucosal epithelium. Most individuals who develop pemphigus are middle-aged and older. There are three major variants:
 - (1) pemphigus vulgaris, (Prototype)
 - (2) pemphigus foliaceus, and
 - (3) pemphigus vegetans.
- ☒ Characterized by ACANTHOLYSIS- separation of epithelial cells from each other = through loss of intercellular substances plus disruption of the DESMOSOMES= so we have individually -separated- smooth surfaced-- epidermal cells.

- ☒ The intercellular space of the macula adherens is conspicuously wider (up to 30 nm) than that of the zonula adherens and is occupied by a dense medial band, the intermediate line. This line represents extracellular portions of transmembrane glycoproteins, the desmogleins and desmocollins, which are members of the cadherin family of Ca^{2+} -dependent cell adhesion molecules. In the presence of Ca^{2+} , extracellular portions of desmogleins and desmocollins bind adjacent identical molecules of neighboring cells (homotypic binding).
- ☒ Vesicle: Fluid-filled raised area 5 mm or less in diameter (papule that contains clear fluid; a blister).
- ☒ Bulla: Fluid-filled raised area more than 5 mm in diameter; a large vesicle.
- ☒ Vesicles and bullae (blisters) can occur as a secondary phenomenon in unrelated conditions
 - a) herpesvirus infection,
 - b) spongiotic dermatitis
 - c) thermal burns
- ☒ Sites of blister formation
 - In a subcorneal blister, the stratum corneum forms the roof of the bulla (as in pemphigus foliaceus).
 - In a suprabasal blister, a portion of the epidermis including the stratum corneum forms the roof (as in Pemphigus vulgaris).
 - In a subepidermal blister, the entire epidermis separates from the dermis (as in bullous pemphigoid and dermatitis herpetiformis).



9.2. Pathogenesis

- ☒ Both pemphigus vulgaris and pemphigus foliaceus are caused by a type II hypersensitivity reaction (antibody directed against a fixed tissue antigen); and show linkage to specific HLA types.
- ☒ Patient serum contain pathogenic IgG antibodies to intercellular desmosomal proteins (desmoglein types 1 and 3) of skin and mucous membranes. The distribution of these proteins within the epidermis determines the location of the lesions. By direct immunofluorescence, lesional sites show a characteristic netlike pattern of intercellular IgG deposits.
- ☒ The antibodies seem to function primarily by disrupting the intercellular adhesive function of the desmosomes and may activate intercellular proteases as well.

9.3. Subtypes

9.3.1. Pemphigus Vulgaris

9.3.1.1. Introduction

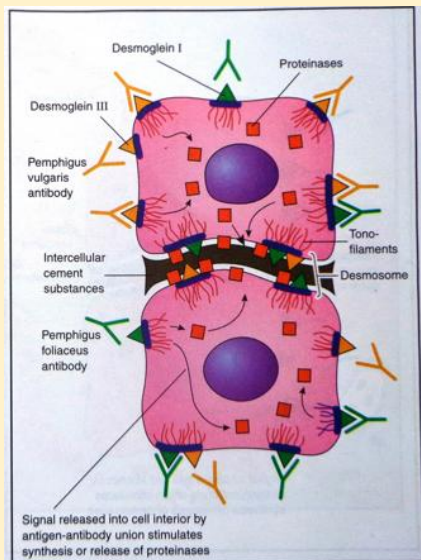
- ↳ Pemphigus vulgaris is a chronic disease characterized by the formation of blisters over large surfaces of skin and oral mucosa. It is mediated by cytotoxic autoantibodies that bind to the surface of epidermal cells, causing acantholysis and intraepidermal fluid accumulation. The autoantigen has been identified as a component of desmosomes, called desmoglein III. Pemphigus vulgaris most often affects middle-aged people but can occur at any age. In approximately 60% of cases, it begins with oral lesions. The skin bullae rupture, and the entire surface may become denuded. Corticosteroids provide relief, but if no treatment is given, the disease may be lethal.
- ↳ Blistering Skin Disorder Caused by Antibodies to Keratinocytes
- ↳ Dyshesive disorders are cutaneous diseases in which blister formation is secondary to diminished cohesiveness between epidermal keratinocytes.
- ↳ Pemphigus vulgaris (PV) (Greek, pemphix, “bubble”) is a chronic, blistering skin disorder that is most common in people between 40 and 60 years of age, but is seen in all age groups, including children. All races are susceptible, but persons of Jewish or Mediterranean heritage are at greater risk.

9.3.1.2. Pathology

- ☒ The blister in PV forms because the outer epidermal layers separate from the basal layer. This suprabasal dyshesion results in a blister that has an intact basal layer as a floor and the remaining epidermis as a roof. Desmoglein 3 is concentrated in the lower epidermis, explaining the location of the blister.

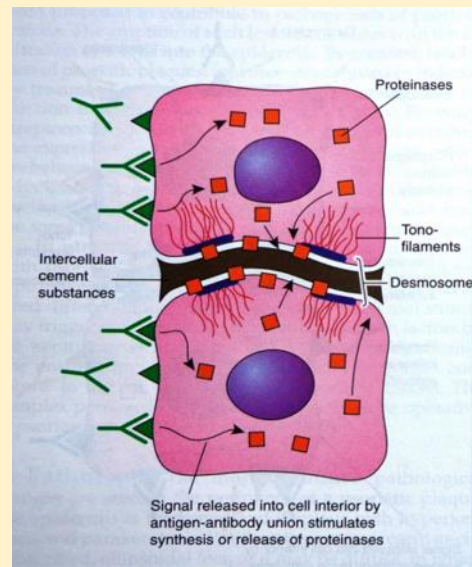
- ⊗ The blister contains moderate numbers of lymphocytes, macrophages, eosinophils and neutrophils. Distinctive, rounded keratinocytes, termed acantholytic cells, are shed into the vesicle during dyshesion. .
- ⊗ The basal cells remain adherent to the basal lamina and form a layer of “tombstone cells.” Dyshesion may extend along dermal adnexa and is not always strictly suprabasal. The subjacent dermis shows a moderate infiltrate of lymphocytes, macrophages, eosinophils and neutrophils, predominantly around the capillary venular bed.

1



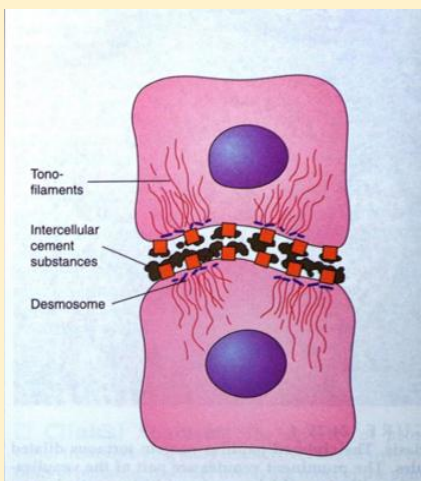
- ***Pemphigus vulgaris. A pathogenic mechanism of suprabasal dyshesion is shown. A circulating autoantibody binds to an antigen on the outer leaflet of the plasma membrane (desmosome) of the keratinocyte, especially in the basal regions.***

2



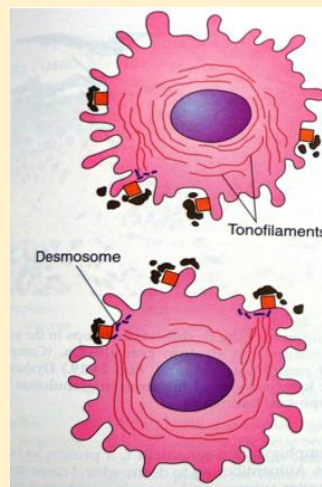
- ***Antigen-antibody union results in release of a proteinase (plasmin).***

3



- ***The proteinase interacts with intercellular cement, initiating dyshesion***

4



- ***Desmosomes deteriorate, tonofilaments clump about the nucleus, the cells round up and separation is complete.***

Other diseases caused by dyshesion that have a pathogenetic mechanism similar to PV include pemphigus foliaceus, pemphigus erythematosis and drug-induced pemphigus (mostly associated with penicillamine and captopril). In pemphigus foliaceus, IgG antibodies to desmoglein 1, a desmosomal protein, cause dyshesion in the outer spinous and granular epidermal layers (in contrast with the suprabasal dyshesion in pemphigus vulgaris). Pemphigus foliaceus and pemphigus erythematosis feature dyshesion in the spinous layer. Paraneoplastic pemphigus has been described in association with cancers, usually lymphoproliferative tumors.

10. Diseases of the Basement Membrane Zone

☒ These include the following

- a. *Epidermolysis Bullosa*
- b. *Bullous Pemphigoid*
- c. *Dermatitis Herpetiformis*
- d. *Erythema Multiforme*
- e. *Systemic Lupus Erythematosus*
- f. *Lichen Planus*

☒ Sites for blister Formation

 In a subepidermal blister, the entire epidermis separates from the dermis (as in bullous pemphigoid and dermatitis herpetiformis).

10.1. Bullous Pemphigoid

10.1.1. Introduction

- ✓ Bullous pemphigoid is a blistering autoimmune disease resembling pemphigus vulgaris. In contrast to pemphigus, it rarely affects oral mucosa.

Bullous pemphigoid is mediated by antibodies to glycoproteins in the lamina lucida of the epidermal–dermal basement membrane. Deposition of antibodies along the epidermal–dermal basement membrane leads to the formation of bullae. Typically such vesicles and bullae are subepidermal (i.e., between the basement membrane and the basal layer of the epidermis that was lifted off the basement membrane).

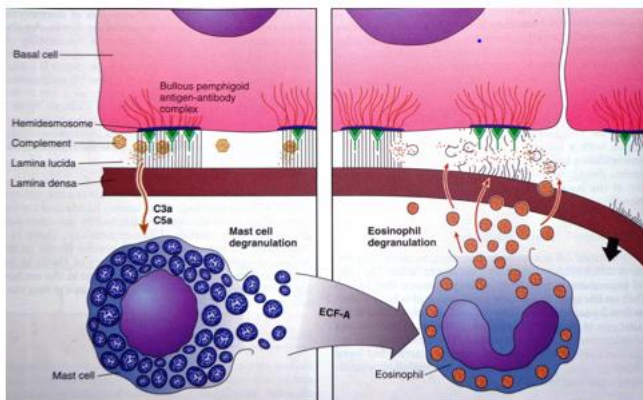
- ✓ Bullous Pemphigoid Is a Subepidermal Blistering Disease Caused by Autoantibodies Against Basement Membrane Proteins
- ✓ BP is a common, autoimmune, blistering disease with clinical similarities to pemphigus vulgaris (thus the term “pemphigoid”) but in which acantholysis is absent. The disease is most common in the later decades of life and shows no predilection regarding race or gender.

10.1.2.

Pathogenesis

1

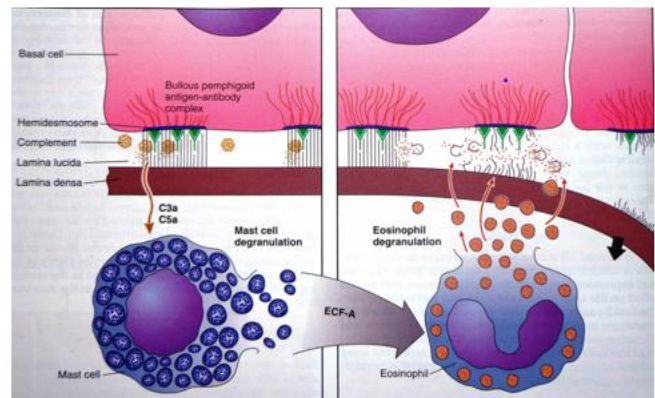
Bullous pemphigoid (BP). Pathogenetic mechanisms of blister formation are outlined. A circulating antibody to an apparently normal glycoprotein—BP antigen—in the lamina lucida precipitates the pathogenetic events in bullous pemphigoid.



3

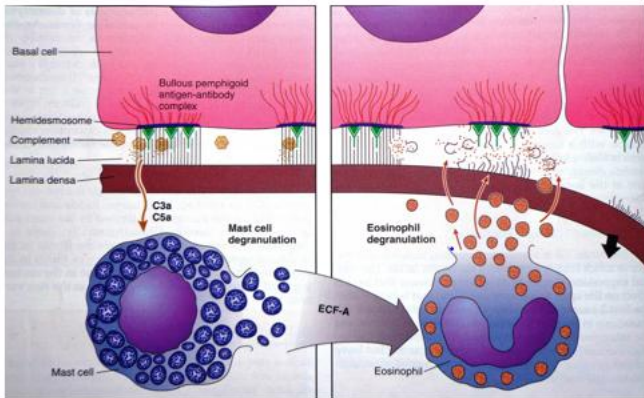
2

Antigen–antibody union activates complement, and the anaphylatoxins C3a and C5a are produced. These degranulate mast cells, resulting in the release of eosinophilic chemotactic factors.

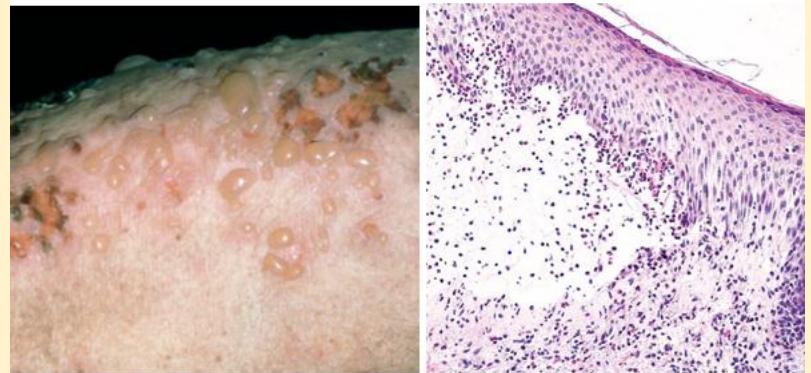


4

The tissue-damaging substances of eosinophilic granules cause vesicle formation at the lamina lucida, with some breakdown of the lamina densa. ECF-A = eosinophil chemotactic factor-A.



- **Bullous pemphigoid.** A. The skin shows multiple tense bullae on an erythematous base and erosions, distributed primarily on the medial thighs and trunk. B. A subepidermal blister has an edematous papillary dermis as its base. The roof of the blister consists of the intact, entire epidermis, including the stratum basalis. Inflammatory cells, fibrin and fluid fill the blister. (Bullous Pemphigoid Is a Subepidermal Blistering Disease Caused by Autoantibodies Against Basement Membrane Proteins)



10.2. Dermatitis Herpetiformis

10.2.1. Introduction

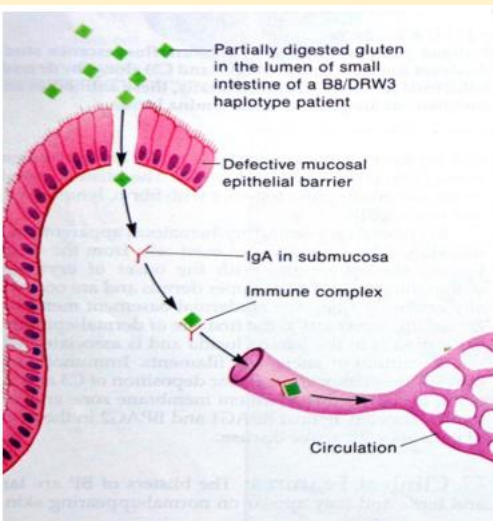
- ✓ Dermatitis herpetiformis is a chronic autoimmune skin disease characterized by recurrent eruptions of itchy vesicles surrounded by erythematous patches.
- ✓ The disease is mediated by deposits of IgA along the epidermal–dermal junction, especially at the tips of dermal papillae.
- ✓ Subepidermal vesicles form and are accompanied by infiltrates of neutrophils in their wall.
- ✓ Most patients also have celiac sprue. Both the intestinal and the skin disease respond favorably to a gluten-free diet.
- ✓ Pruritic, symmetric, grouped vesicles on an erythematous base are seen on the elbows and knees.
- ✓ Dermatitis Herpetiformis Reflects Gluten Sensitivity and Immune Complex Deposition
- ✓ Dermatitis herpetiformis (DH) is an intensely pruritic cutaneous eruption characterized by urticaria-like plaques and small subepidermal vesicles over the extensor surfaces of the body.

10.2.2. Pathology

- ✓ This disease results from formation of antibodies against gliadin, a protein found in the gluten fraction of wheat. Patients with dermatitis herpetiformis may develop antibodies of the IgA and IgG classes to gliadin and reticulin, a component of the anchoring fibrils that tether the epidermal basement membrane to the superficial dermis.

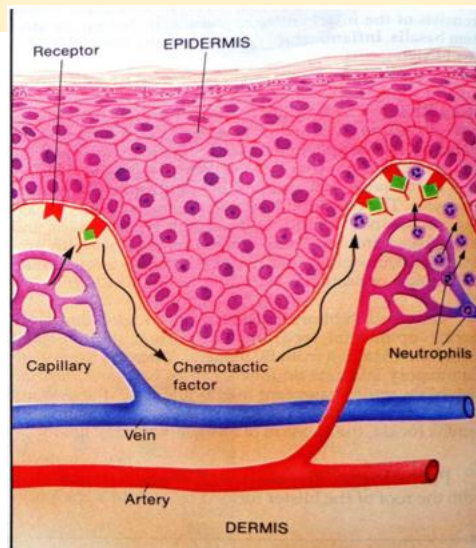
- ✓ In addition, individuals with certain histocompatibility types (HLA-B8 and HLA-DRw3) are particularly prone to this disease. It is thus thought that genetically predisposed persons may develop IgA antibodies to components of dietary gluten and that these antibodies (or immune complexes) then crossreact with or are deposited in the dermal papillae of the skin, resulting in clinical disease. Some individuals with dermatitis herpetiformis and enteropathy respond to a gluten-free diet (as with celiac disease).
- ✓ Celiac disease, also known as gluten-sensitive enteropathy, is the prototype of a noninfectious cause of malabsorption resulting from a reduction in small intestinal absorptive surface area.
- ✓ Celiac disease is believed to be quite common, affecting 1 in 300 persons both in Europe and in the United States, and many patients have subclinical disease.
- ✓ The basic disorder in celiac disease is immunological sensitivity to gluten, the component of wheat and related grains (oat, barley, and rye) that contains the water-insoluble protein gliadin.

1



Formation of immune complexes in submucosa of small intestine. Passage of immune complexes into circulation.

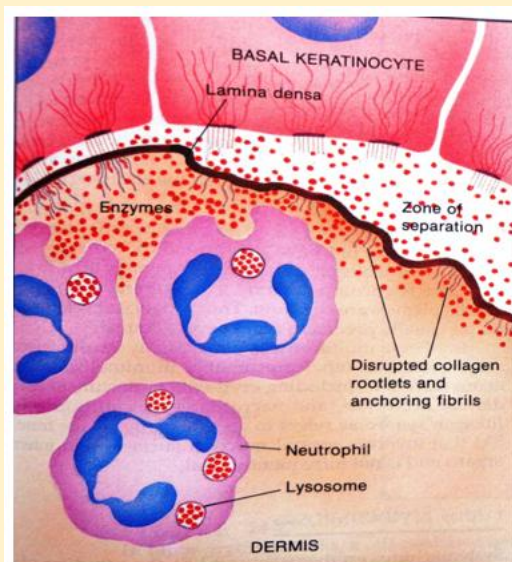
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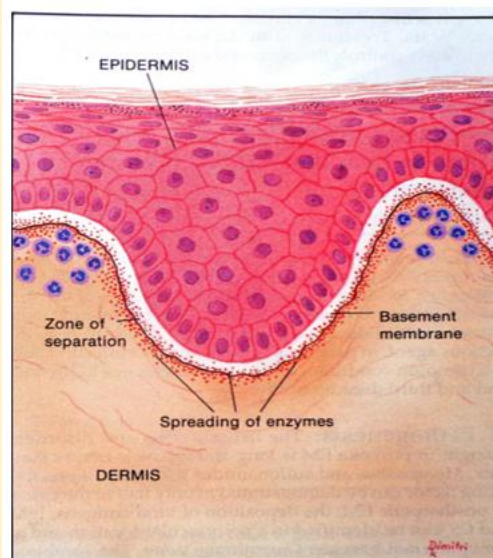
• **Receptor-immune complex union releases neutrophil chemotactic factor. Neutrophils migrate to the tips of the papillae**

3

4



Dissolution of basal rootlets and anchoring fibrils by enzymes released by neutrophils. Early dermo-epidermal separation.



Concentration of neutrophils at the tips of the papillae. Spreading of enzymes along basement membrane. Lifting away of lamina densa.

11. Albinism

⊗ *Three types of congenital hypopigmentation occur:*

⊗ **Generalized or oculo-cutaneous albinism**

⊗ **Ocular albinism**

⊗ **Partial albinism**

⊗ **Generalized or oculocutaneous albinism** is recessively inherited and shows generalized absence of pigmentation in the skin and hair, hypopigmented ocular fundi, translucent irides, and nystagmus.

⊗ Occasionally generalized albinism is associated with congenital deaf-mutism.

⊗ In generalized albinism silver stains show absence of melanin in the epidermis.

⊗ Melanocytes are present in normal number.

⊗ **In partial albinism** the depigmented areas show absence of melanin.

⊗ On examination no melanocytes are found.

12. Primary Neoplasm's of the Skin.

- Benign Keratinocytic neoplasm's related to Human Papilloma Virus infection.
- Benign Keratoses.
- Squamous cell Carcinoma.
- Basal cell Carcinoma. = Rodent Ulcer
- Merkel cell Carcinoma. (i.e. Neuroendocrine cell carcinoma of the skin).

NOTE:

Neoplasms arising from the cell – types found in the EPIDERMIS layer of the skin
MALIGNANT TUMOURS OF THE EPIDERMIS INCLUDE the last three above

12.1. Benign Keratinocytic neoplasms related to Human Papilloma Virus infection.

- ✓ Human Papilloma Viruses: DNA- viruses- whose infection can induce development of skin tumours. The lesions are circumscribed; elevated above the skin and often papillary. Commonly called Viral Warts/ VERRUCAE.
- ✓ There are several clinico- pathological types of warts.
- ✓ Classified, based on organ/ site of infection; and appearance of the lesions

12.1.1. Verrucae vulgaris (vv)

- circumscribed, firm, elevated growth having a papillomatous, hyperkeratotic surface. Vv may occur anywhere on the skin (in rare instances – on the oral mucosa). The fingers – most common site of vv.
- shows acantosis, papillomatosis, and hyperkeratosis.
- Typical appearance of a verruca. The histologic features include papillomatosis, acanthosis, hyperkeratosis with parakeratosis and elongated rete ridges appearing to point towards the centre. Foci of vacuolated cells (koilocytes) are found in the upper stratum malpighii.
- Inset shows koilocytes and virus-infected keratinocytes containing prominent keratohyaline granules.
- Groups of large vacuolated cells lie in the upper stratum malpighii and in the granular layer. The thickened horny layer contains many parakeratotic cells.
- Squamous epithelial-lined fronds have fibrovascular cores. The blood vessels within the cores extend close to the surface of verrucae, which makes them susceptible to traumatic hemorrhage.

12.1.2. CONDYLOMA ACCUMINATUM:

- ✓ Primary genital warts; very large; associated with sexually transmitted disease(s). H.P.Virus – serotypes 6 and less commonly 11.
- ✓ Squamous carcinoma may develop in these lesion; especially giant condyloma accuminatum; where HP Virus serotypes – 16/ and 18 are identified similar to C.I.N
- ✓ There are parakeratosis and tremendous acanthosis. The cells of the stratum malpighii show marked intracellular edema and round hyperchromatic nuclei. In the dermis, numerous dilated blood vessels and dense chronic inflammatory infiltrate are present.

12.2. Benign keratoses of the skin

- Keratosis - any horny/ hard growth or callosity composed of benign; localised--proliferation of keratinocytes.
- Three of these conditions are common;
 - a. Seborrheic keratosis (these are scaly lesions; frequently pigmented; as elevated plaques or papules; the scales can easily be rubbed off.)
 - b. Actinic Keratosis
 - c. Keratocanthoma

12.2.1. Seborrheic Keratoses:

- these are scaly lesions; frequently pigmented; as elevated plaques or papules, the scales can easily be rubbed off.
- Most common keratoses;
- aetiology is unknown. Similar to Basal cell papilloma - a benign tumor in elderly patients; in later years of life and may be familial.
- **Histologically**: anastomosing cords of proliferating squamous cells with HORNY cysts(i.e. small cysts of keratin).
- Very bland and innocuous but cosmetically a nuisance.
- The lower border of the tumor in general follows a straight line that could be drawn from the normal epidermis at one end of the tumor to the normal epidermis at the other end.
- Thick interwoven tracts of epidermal cells compose the tumor. Most of the epidermal cells have appearance of epidermal basal cells. They referred to as

basaloid cells. Interspersed are cystic inclusions of horny material caused by invagination of the horny layer. They are referred to as pseudocyst.

12.2.2. Actinic Keratosis.

- A keratosis which develops in sun-damaged skin (i.e. damage due to sun rays).
- usually on Backs of Hands or the FACE
- Epidermis is thinned out; is parakeratotic at the top; but underlying basal keratinocytes are dysplastic (i.e.—cytologically atypical).
- Usually is considered as a pre-malignant condition; with the passage of time it may develop into--- Full blown--- INVASIVE squamous cell carcinoma.
- A low-power view reveals cytological atypia within the stratum basalis and lower stratum spinosum, with loss of polarity. A bandlike, lymphocytic infiltrate is frequently present. Parakeratosis is present here only in small focus.
- High power examination of an actinic keratosis reveals striking cytological atypia of the basal keratinocytes, the hallmark of actinic keratosis.

12.2.3. KERATOACANTHOMA

- is rapidly growing keratotic lesion on sun-exposed skin within 3-6 wks; as a volcano- like nodule with central umbilication
- can grow to 2-3cm in diameter; rare variants can reach size of 15-20 cm; usually they regress spontaneously in 6- 12 months -> leaving an atrophic scar.
- this lesion is considered as a type of squamous cell carcinoma, which heals spontaneously by itself in the majority of patients.
- **Histologically:** as a keratin- filled “crater”- which is lined by proliferating keratinocytes; it invades deep into the Reticular dermis; with overhanging edges on both sides: plus marked inflammatory infiltrate in the dermis.
- Low magnification. There is large, central, keratin-filled crater. On the right, the epidermis extends like a lip over the side of the crater. Irregular proliferation of the epidermis extend into the dermis at the base of the crater.
- High magnification of the epidermal proliferations at the base of the crater shows their resemblance to squamous cell carcinoma. However, there is more keratinization than is usually seen in squamous cell carcinoma, giving the tumor islands a glassy appearance.

12.3. SQUAMOUS CELL CARCINOMA

- A malignant tumour, arising from epidermal Keratinocytes/squamous cells;- the malignant cells typically resemble differentiated Keratinocytes.
- Is second only to Basal cell carcinoma in incidence.
- Predisposing core factors-:[in the cause of these cancers;]
 - a) Ultraviolet light rays
 - b) Ionizing radiation
 - c) Some chemical carcinogens e.g. soot/carbon particles in chimney sweepers---: in the U.K. -: in the 1700's -1800's AD = scrotum-squamous cell carcinomas.
 - d) Human papilloma virus infection i.e. Vulval squamous cell carcinoma + V.I.N-I/II/III.
- An ulcerated, encrusted, and infiltrating lesion is seen on the sun-exposed dorsal of a finger
- The entire epidermis is replaced by atypical keratinocytes. Mitoses and multinucleation of keratinocytes are apparent, as is apoptosis.
- invasive component. High-power view reveal irregularly shaped lobules of strikingly atypical keratinocytes that have invaded to the level of the midreticular dermis.

➤ HISTOPATHOLOGY:

- ✓ Similar to squamous cell carcinomas in other organs e.g. cervix, lung
- ✓ Malignant cells look like or mimic the squamous cells in the epidermis, with-
DESMOSOMES/ intracellular bridges , keratin formation/ dyskeratosis/
Keratohyalin pearls formation PLUS invasion through basement membrane into the dermis.
- ✓ NOTE;
- ✓ Well differentiated
- ✓ Moderated differentiated
- ✓ Poorly differentiated

12.4. BASAL CELL CARCINOMA

- Arises from the Pluripotent cells in the basal layer of the epidermis= stem-basal cells

- Is a locally invasive tumour, and the tumour cells resemble Basal Keratinocytes
 - Most commonest tumour in persons with a pale skin, i.e. caucasians, may be locally aggressive but METASTASIS does not occur usually, if it does, then this is very extremely-rare.
 - **Pathogenesis:** usually develops on sun-damaged skin== on patients with FRECKLES OR THOSE WITH FAIR SKIN.
 - Rarely---- arises from== non-sun damaged skin,- face,- dorsum of hands.
 - Is a component of a number of hereditary syndromes.
 - In which case, it arises from non-sun damaged- skin=sites
 - BASAL CELL CARCINOMA of the skin overlying the nose has ulcerated and invaded deep tissues. Microscopic examination shows a sclerosing and infiltrative lesion. Irregularly branching strands of tumor cells permeate the dermis, with induction of a cellular, fibroblastic, hyaluronic acid-rich stroma.
 - **Fibrosing basal cell carcinoma.** Innumerable small groups of closely packed tumor cells, many of them arranged in elongated strands are embedded in a dense fibrous stroma.
 - CLINICO-HISTOPATHOLOGICAL FEATURES -: (gross and microscopic)
- ✓ there are different sub-types of Basal Cell carcinoma (B.C.C)

1) Superficial – B.C.C

2) Pigmented -- B.C.C

- ✓ In superficial – B.C.C = multiple nests/islands of basal-cells, deeply basophilic oval/round , small cells = attached to the epidermis= plus invading the papillary dermis. At the periphery of the cords/islands= is a palisade-: of columnar-like Keratinocytes.
- ✓ In superficial type. Buds of atypical basaloid keratinocytes extend from the overlying epidermis into the papillary dermis. The peripheral keratinocytes mimic the stratum basalis by palisading. The separation artifact is present because of poorly formed basement membrane components and the hyaluronic acid-rich stroma that contains collagenase.
- ✓ **Clinical features of Basal Cell carcinoma-:** (i.e. used also--- to sub-type the === B.C.Ca's)
- ✓ may present as-:

- i. Pearly-PAPULE = 2-3mm in size, covered by stretched epidermis (Early Lesion)
- ii. Rodent ulcer= as a small ulcerative , crater--- in the centre-- of papular lesion.
- iii. As an ill defined- pale= firm , scar-like tumour = in the cases of desmoplastic sub-types of B.C.C.
- iv. Pigment –BCC-- subtypes-: grossly resembles Malignant Melanoma-: due to the excess Melanin pigmentation.

NOTE: NEVO-Basal Cell Carcinoma Syndrome =: is the occurrence of Multiple tumours-- in the cortex of a complex multisystem disease.

For example

- ✓ In some patients with Basal cell carcinomas; have mandibular cysts + Dyskeratoses in the palms and soles Plus a predisposition to other types of neoplasms.
- ✓ The B.C.Ca's--- in these syndromes--- occur at very younger ages-> and may number x10 to over 100 of them ==involving both-- sun exposed-skin areas and non-sun exposed areas.

12.5. Malignant melanoma (MM).

- Malignant melanoma, the most malignant of the cutaneous neoplasm, arises from the epidermal melanocytes.
- It may arise de novo or in a preexisting nevus.
- Malignant melanoma is rare before puberty, but fatal cases have been reported in children.

12.5.1. Dyplastic Nevi

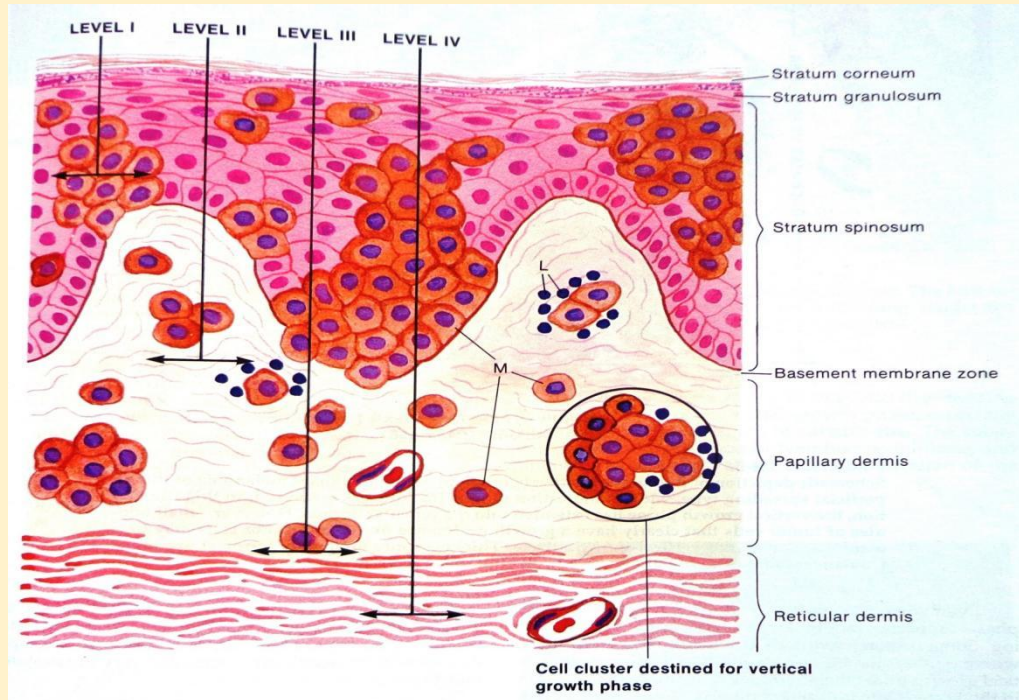
- **Dysplastic nevi** are pigmented lesions that clinically and histopathologically differ from regular nevi and thus appear atypical. This term is poorly defined and often misused, and a National Institutes of Health consensus conference recommended that the term dysplastic be abandoned and replaced by atypical nevus. Nevertheless, some facts deserve to be mentioned.
- Atypical nevi may give rise to melanoma in situ or invasive melanoma. This transition can be recognized with certainty only by microscopic examination of biopsied lesions. Accordingly, any suspicious mole, especially those larger than 6 mm in diameter, must be removed and examined microscopically.
- Clinical features suggesting an atypical nevus are their size (>6 mm), irregular borders, and variegated color (shades of brown, tan, and light red). A “fried egg” appearance (dark center surrounded by less pigmented rim that has indistinct margins) is often cited as typical for these atypical nevi.

12.5.2. Types of Malignant Melanoma

a) Superficial spreading

- ✓ Melanocytes grow singly within the epidermis at all levels and as large, irregularly-sized nests at the dermal-epidermal junction. Tumors cells are present in the papillary dermis

b) Nodular



- ✓ Intraepidermal growth is essentially absent. The tumor expands the papillary dermis and distorts the reticular dermal junction

c) Lentigo maligna melanoma (Hutchinson's freckle)

- ✓ is a large pigmented macule that occurs in older people or in sun-damaged skin.
- ✓ Atypical melanocytes grow largely at the dermal-epidermal interface, with extension down the external root sheath of follicles.

d) Acral lentiginous melanoma

- ✓ presents on the palms, the soles, or the genital or oral mucosa as a pigmented macule that that may be deceptively bland and lentigo-like at first but eventually becomes invasive.
- ✓ Clinical appearance of the sole of the foot in a patient with malignant melanoma of the acral lentiginous type.

- Superficial spreading melanomas account for 75% of all MMs diagnosed in the United States. Nodular melanomas (15%), lentigo maligna (5%), and acral lentiginous melanoma (2%–3%) are less common.

- Superficial spreading and Nodular types occur on any part of the body but most often on the legs, shoulders, and upper back.
- The lesions presents as a gradually enlarging pigmented nodule surrounded by erythema. The nodule may show crusting, bleeding or ulceration.

13. Inflammation

- Inflammation is the directed tissue and its microcirculation response to noxious and injurious external and internal stimuli.
- An inflammatory response is a reaction to cell injury that can occur only in vascularized tissues of multicellular organisms.

13.1. Etiology (causes) of inflammation

- ✓ Infective Bacteria, viruses, parasites and their metabolic by-products.
- ✓ Immune response
- ✓ Hypersensitivity, immune complex, or autoimmune reactions.
- ✓ Physical factors
- ✓ Trauma, heat (cold), electricity, radiation.
- ✓ Toxic chemical agents.
- ✓ Other circumstance leading to tissue damage (e.g. vascular or hormonal disturbance).

13.2. Types of inflammation.

- Acute inflammation.
 - ✧ Is an immediate reaction to injury. Typically it has sudden onset and is of short duration. It lasts a few hours or days.
 - ✧ Acute inflammation is typically mediated by neutrophils.
 - ✧ Main events in acute inflammation are:
 1. Hemodynamic changes (vasodilatation, which causes local heat and redness).
 2. Increased vascular permeability, which causes local edema and swelling.

3. Emigration of leucocytes from blood vessels into the tissues:

- Influx of neutrophils
 - Influx of monocytes-macrophages
 - Phagocytosis
 - Release of intracellular enzymes from phagocytes
- Subacute inflammation.
 - Chronic inflammation.

13.3. Dermatitis (eczema).

- Dermatitis – an inflammatory condition of the skin caused by noxious and injurious external and internal agents.
- The term dermatitis and eczema now are used generally as synonyms. They refer to an inflammation of the skin based on an allergic response of the skin to a variety of agents. Such as chemicals, proteins, bacteria and fungi.
- The exciting allergen may act on the skin from either the outside or the inside.
- The term dermatitis is one of the most misused in dermatopathology, for variety of reasons. However, this is incorrect and the term should be restricted to a specific type of epidermal reactions characterised by mixture of focal parakeratosis, hyperkeratosis, acanthosis, and most importantly, spongiosis.

13.4. Bacterial disease of the skin

13.4.1. Impetigo infectiosum.

- May be caused by streptococci or staphylococci.
- Bullae are primary lesion in either case.
- In the streptococcal variety the bullae soon rupture, leaving sharply demarcated erosion that becomes covered with heavy yellowish crust.
- In the more common staphylococcal variety the clinical appearance may be the same as that in streptococcal type, but in some cases the bullae are more durable and dominate the clinical picture.

- Staphylococcal impetigo may occur in the newborn as an extensive bullous eruption, called Ritter's disease.

Histopathology.

- In both the streptococcal and the staphylococcal varieties of impetigo the bulla arises directly beneath the horny layer.
- The bulla contains a lot of neutrophils. Only rarely it is possible to recognize with Giemsa's or Gram's stain groups of cocci within the bulla. They may lie extracellularly or within neutrophils.
- Occasionally, a few acantholytic cells can be observed at the floor of the bullae as a result of the proteolytic action of the neutrophils.
- The stratum malpighii underlying the bulla shows spongiosis; neutrophils can be seen migrating through it. The upper dermis contains moderately severe inflammatory infiltrate of neutrophils and lymphocytes.
- At later stage, when the bulla has ruptured, the horny layer is absent, and a crust composed of fibrin and nuclear debris of neutrophils may be seen covering the stratum malpighii.

13.4.2. Acute deep folliculitis (furuncle).

- A furuncle, caused by staphylococci, consists of a tender, red perifollicular swelling terminating in suppuration and necrosis.

Histopathology.

- A furuncle presents a perifollicular abscess composed mainly of neutrophils. The hair with its follicular wall and sebaceous gland is destroyed in the process.

13.4.3. Erysipelas.

- An acute superficial cellulitis of the skin caused by streptococci.
- Characterized by the presence of well-demarcated slightly indurated, dusky-red area having an advancing, palpable border.
- Occasionally, vesicles or bullae are present.

- In certain individuals erysipelas has tendency to recur periodically in the same area.

Histopathology.

- The dermis shows marked edema and dilatation of the lymphatics and capillaries.
- When the edema is intense, subepidermal bullae may be present.
- There is diffuse infiltrate composed chiefly of neutrophils and some lymphocytes.
- In cases of recurring erysipelas the lymph vessels of the dermis and subcutaneous tissue shows fibrotic thickening of their walls with partial or complete occlusion of the lumen.

13.4.4. Hidradenitis suppurativa.

- Represents a chronic suppurative and scarring process affecting the apocrine glands of the axillary and the genito-anal region.
- Early lesion consist of red tender nodules that become fluctuating and finally discharge pus.
- Healing take place with formation of sinus tracts and with considerable scarring.

Histopathology.

- The earliest cellular reaction occurs within and around the apocrine glands.
- The infiltrate at this stage composed predominantly of neutrophils. And abscess formation results.
- Later on lymphocytes and plasma cells predominante, and foreign-body giant cells are present near the remnants of hair follicles.
- In the stage of healing, deep, tortuous invaginations of the epidermis filled with keratin are seen, representing the sinus tracts.

13.4.5. Pyoderma gangrenosum.

- The lesion of pyoderma gangrenosum, of which there may be one or several, start with a small dermal abscess that breaks through the epidermis, causing an ulcer.

- The ulcer spreads peripherally. The skin at the edge of the ulcer is purplish red, edematous and undermined.
- Pyoderma gangrenosum is associated with ulcerative colitis (not infrequently).

Histopathology.

- The histologic appearance is not diagnostic. In the region of the ulcer the epidermis is absent.
- The upper dermis shows necrosis and is permeated by an acute inflammatory infiltrate.
- Farther down, the infiltrate is chronic inflammatory in nature, consisting of lymphocytes, plasma cells, histiocytes and fibroblasts. Frequently, a few foreign-body giant cells are present.
- The number of blood vessels is increased.
- The cellular infiltrate may extend deeply into subcutaneous tissue.
- Fibrosis occurs in areas of healing.
- Often, the epidermis at the edge of an ulcer shows considerable proliferation, so that the picture of pseudocarcinomatous hyperplasia may result.
- Intra-epidermal abscesses may occur in this region.

13.4.6. Tuberculosis of the skin (lupus vulgaris)

- The lesions are found most commonly on the face and consist of well-demarcated, red-brown patches. Superficial ulceration occurs occasionally.

Pathology

- Tubercles composed of epithelioid and giant cells and giant cells are present.
- Caseation and other forms of necrosis are almost always absent.
- The upper portion of the dermis may show extensive inflammation.
- Occasionally the infiltrate can extend into the subcutis, destroying the cutaneous appendages. The acid-fast bacilli may be difficult to demonstrate by special staining methods.

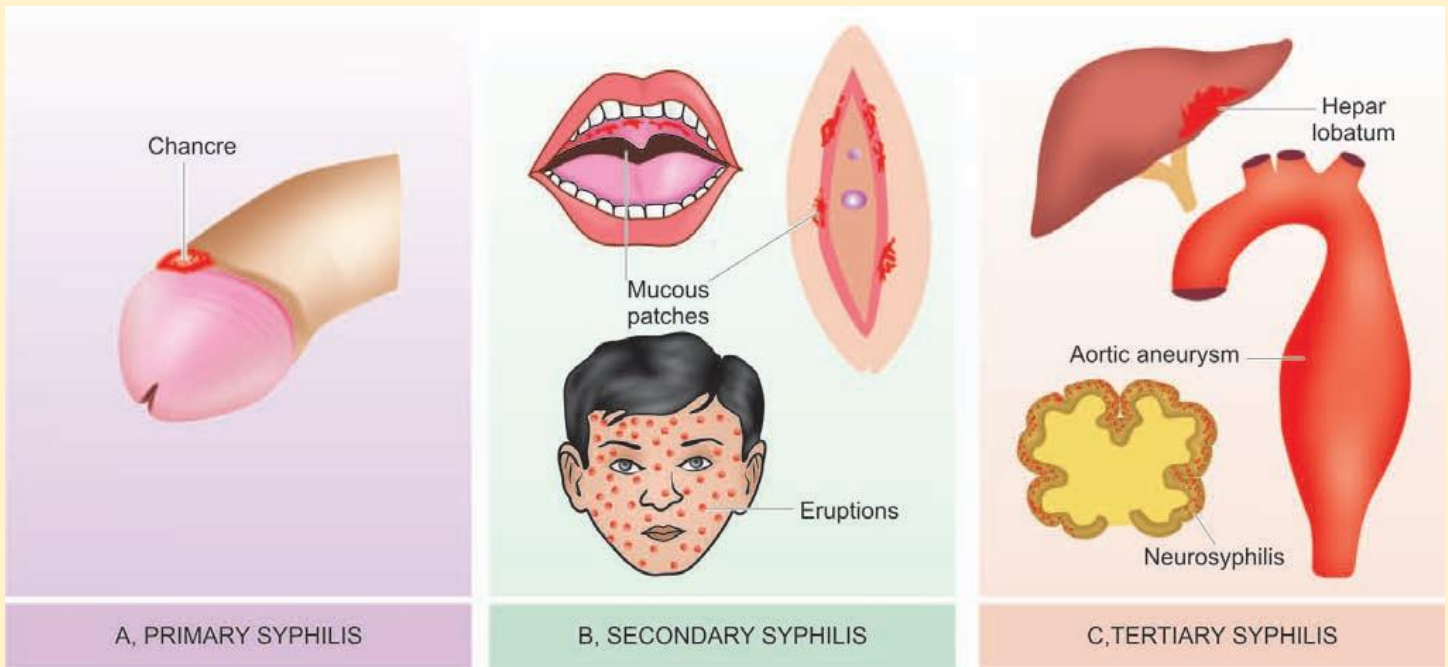
13.4.7. Leprosy.

- A slowly developing disease caused by *Mycobacterium leprae*.

- Althou it manifests first in the skin and peripheral nerves, the bacterium also may be found in parenchymal organs.
- Typically, the patient has hypoesthesias and hypopigmented papules over the skin of the face, ears, neck and trunk.
- There are several forms of disese; the form that develops in part on host resistance to the infecting organisms.
- In the **lepomatous form**, host defense is inadequate, and bacterial multiplication is uncontrolled. The dermis shows diffuse infiltration by foamy histiocytes that are full of organisms.
- In the **tuberculoid form**, there is strong resistance to the infection. There are epithelioid granulomas that contain no organisms.
- **Tuberculoid leprosy.** (strong resistance to the infection) The infiltrate consists of epithelioid cell tubercules, showing a slight admixture of lymphocytes, particularly at their mardins. Thus the histologic picture is very much like that of sarcoidosis.

13.4.8. Syphilis.

- Caused by the spirochete *Treponema pallidum*.
- Transmitted either by venereal contact or transplacentally by an infected mother to her unborn child (congenital syphilis).
- Three stages of the venerally transmitted disease are recognized:



1. Primary syphilis.

- ✓ characterized by the development of a painless, hard-based ulcer, the chancre, at the point of inoculation, usually in the genital area.
- ✓ Primary lesion is 'chancre' on glans penis
- ✓ This stage often is asymptomatic. A chancre is a grossly shallow ulcer.
- ✓ **Microscopically**, the epithelium is ulcerated, and the ulcer base shows an intense, predominantly plasmacytic infiltrate, accompanied by an obliterative endarteritis.

2. Secondary syphilis.

- ✓ usually occurring 1 to 4 months later.
- ✓ Mucocutaneous lesions—mucous patches on oral and vaginal mucosa and generalised skin eruptions.
- ✓ Characterized by diffuse skin rash and generalized (especially inguinal) lymphadenopathy, often accompanied by fever and malaise.
- ✓ The rash may develop into condyloma lata.

- ✓ Skin biopsies of the rash of the secondary syphilis show subepidermal mononuclear-cell (plasmatic) infiltrate, chiefly perivascular; spirochetes may be found.
- ✓ The lymph nodes demonstrate an extensive capsular fibrosis and plasmatic perivasculitis.

3. Tertiary syphilis.

- ✓ Often often appearing after latent periods of 1 to 30 years.
- ✓ Affects the cardiovascular system and central nervous system.
- ✓ Other areas, especially the liver, bones, and joints, may be affected by gummas.
- ✓ The diagnosis of syphilis is made by darkfield examination of material from the primary chancre (when present) or by serologic test.

14. Nonmalignant pigmented lesions.

14.1. Freckle (ephelis)

- is a pigmented macule composed of normal melanocytes, which overreact to sunlight by producing melanin at a rapid rate. The number of melanocytes is not increased.
- Macule: Flat, circumscribed area of any size distinguished from surrounding skin by coloration.

14.2. Lentigo

- a pigmented lesion (few millimeters in diameter) containing an increased number of melanocytes with a hyperpigmentation of adjacent keratinocytes.
- In contrast to ephelis, lentigo does not darken upon exposure to sunlight.
- Histologically the involved skin shows elongation of rete ridges, thus bringing the pigmented cells closer one to another.

14.3. Melanocytic nevus (the common mole)

- a benign neoplasm composed of nevus cells.

- May be congenital or acquired. Congenital nevi are found in approximately 1% of all newborns.
- Acquired nevi appear later, usually at puberty, and involute spontaneously in old age.
- May appear flat, papillomatous, or pedunculated, and it may show marked hyperpigmentation.
- Nevi are classified according to the location of melanocytes (nevus cells) into three groups:
 - a) Junctional nevus.
 - ✓ In clinical appearance, lesions are small, relatively flat, symmetric, and uniform.
 - ✓ On histologic examination, junctional nevi are characterized by rounded nests of nevus cells originating at the tips of rete ridges along the dermoepidermal junction.
 - b) Compound nevus.
 - c) Intradermal nevus.
 - ✓ The upper dermis shows nests and cords of nevus cells, some of which contain varying amounts of melanin. In the lower dermis, the nevus cells are scattered and embedded in collagenous tissue.
 - ✓ The epidermis may show changes, such as papillomatosis and hyperkeratosis, or it may appear normal.

Melanoma

A: Asymmetry

B: Border (irregular)

C: Color changes

D: Diameter > 6 mm (size of a pencil eraser)

E: Evolving (any change in characteristics such as size, shape, color, elevation, new symptoms such as bleeding, itching, or crusting)

Melanomas are more likely to occur in older people, but are also found in younger people.

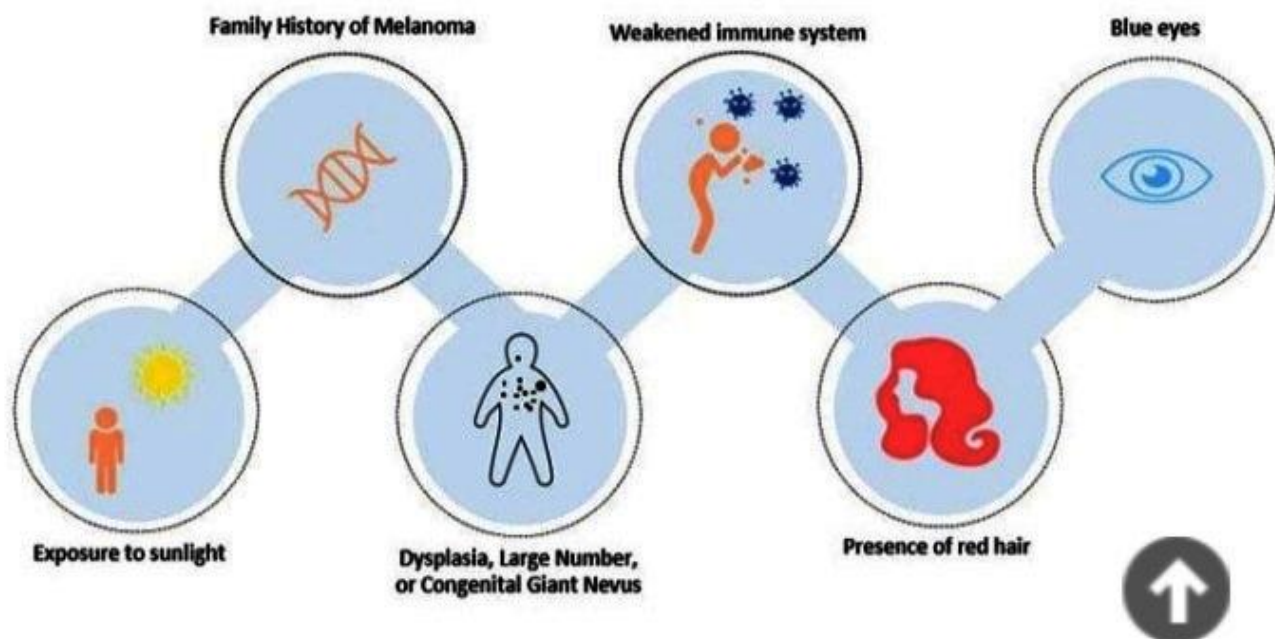


Figure 2 Risk factors of melanoma

People with weak immune systems are more likely to develop melanoma. For example, People who receive organ transplants and are infected with HIV tend to have weaker immune systems and a higher risk of melanoma.

Ultraviolet radiation is closely related to the development of melanoma. It was found that UVB acted on the pyrimidine ring of DNA, causing mutations in the pyrimidines ^[3], thereby damaging DNA. UVA leads to damage and fracture of DNA through the production of reactive oxygen species (ROS), and finally produces gene mutation ^[4].

Skin cancer begins when ultraviolet light damages the DNA of genes that control the growth of skin cells.

At present, the etiology of melanoma has not been fully elucidated, and the main cause of malignant melanoma is the uncontrolled and abundant proliferation of melanocytes.

The occurrence of melanoma is the result of multiple factors such as genes, environment and gene/environment. The main risk factors for melanoma include the following:

Melanoma originates in the neural crest melanocytes. Melanoma forms when melanocytes become malignant. Melanoma is the most deadly malignant tumor in skin cancer ^[1]. It is often found in the skin, but also occurs in eye, nasal cavity, anal rectum, lymph node and other tissues.

Melanoma occurs in the skin and is called cutaneous melanoma. Originating from the eyes, nasal cavity, digestive tract, lymph nodes, vulva, meninges, etc., it is called mucosal melanoma. Melanoma is characterized by high metastasis, high malignancy, high mortality, poor prognosis and poor therapeutic effect ^[2].

Clark levels of skin cancer. In Clark Level I, the cancer is in the epidermis only. In Clark Level II, the cancer has begun to spread into the papillary dermis (upper layer of the dermis). In Clark Level III, the cancer has spread through the papillary dermis into the papillary-reticular dermal interface but not into the reticular dermis (lower layer of the dermis). In Clark Level IV, the cancer has spread into the reticular dermis. In Clark Level V, the cancer has spread into the subcutaneous tissue.

Classification

At present, the clinical tissue classification of melanoma adopts Clark classification ^[11] to classify melanoma into 5 grades according to the depth of invasion. The higher the grade, the worse the prognosis.

Clark I: the tumor is confined to skin layer (in-situ melanoma).

Clark II: tumor cells break through the basement membrane and invade the dermal papilla.

Clark III: the tumor is full of dermal papilla, but not to the dermal reticular layer.

Clark IV: tumor infiltrating dermal reticular layer of collagen fibers.

Clark V: tumor infiltrating subcutaneous tissue.

